Daniel Zhao

danielzhao02@gmail.com | https://astradzhao.github.io/ | https://www.linkedin.com/in/danzhao02/

Education

University of California, San Diego | San Diego, CA — SEP 2024 - DEC 2025

M.S. in Computer Science (AI focus) **GPA:** 3.90

Relevant Coursework: Deep Learning, Machine Learning, NLP, ML Systems, Artificial Intelligence, Recommender Systems, Computer Arch

University of California, Berkeley | Berkeley, CA — AUG 2020 - MAY 2024

B.A. in Computer Science & B.A. in Applied Math **GPA:** 3.90

Professional Experience

Teaching Assistant | ***CSE 234: ML Sys, CSE 253: ML for Music, CSE 258: Recommenders @ UCSD*** — DEC 2024 - PRESENT

- Constructed 3 brand-new assignments for students covering mat-mul optimization, backprop graph construction, speculative decoding
- Hosted 50+ office hours, proctor exams, hold discussion sections, maintain Ruby website under Prof. Hao Zhang & Prof. Julian McAuley

Machine Learning Engineer Intern | ***Qualcomm*** — JUN 2025 - SEP 2025

- Architect **multimodal RAG** system through Langchain, using ONNX-based CLIP embeddings, FAISS vector store, Qwen2.5VL
- Deploy system on Qualcomm-native CPU machine, benchmarking against CUDA host on Haystacks using LLM-as-a-judge
- **Cross-compile Llama.cpp** for Adreno GPU and quantize QwenVL model to Q4_0, achieving over **10x speedup** from baseline CPU
- Extensively profile and fix CLIP projection encoding bottleneck in Llama.cpp C++ binaries, achieving 5x speedup from naive solution

Research Intern | ***BergLab, Hao AI Lab @ UCSD*** — APR 2024 - PRESENT

- Explore interpretability of music and guided real-time music generation through recursive feature machines and sparse autoencoders
- Configured new CLLM models to be compatible with **qLoRA fine-tuning** on Jacobi trajectories, decreasing memory usage over 3x

Machine Learning Engineer Intern | ***Accretional*** — JUN 2024 - SEP 2024

- Designed and implement **RAG vector** embedding model to ground LLM with respect to cloud infrastructure
- Created command-line tool in Golang for indexing and vectorization of files, **semantic & lexical search** of keywords
- Fully constructed VSCode application in Svelte and TypeScript to help users easily implement and deploy APIs in less than 5 seconds

Software Engineer Intern | ***Hinge Health*** — MAY 2023 - AUG 2023

- Designed 5+ endpoints and event consumers in TypeScript and Ruby on Rails, restructuring and building new GraphQL queries
- Presented solutions for over 10 UI/UX features in React, linked repos with RabbitMQ event emitting, deployed in Kubernetes

Full Stack Intern | ***Cisco Meraki*** — MAY 2022 - AUG 2022

- Spearheaded a full stack project to create uplink ping stat graphs, helping **500,000 customers** visualize wireless trends over time
- Collected ping stats data from over 4 million wireless internet nodes, using Scala scripts to build a grabber and Jenkins to test

Research & Other Projects

Muras, an open-source Python framework for multimodal RAG benchmarking and evaluation — OCT 2025 - PRESENT

Steering Autoregressive Music Generation with RFMs - submission to ICLR 2026 — OCT 2025

- Interpretable classification of music using **recursive feature machines** (based on kernel ridge regression & AGOP) as classifiers
- Inject direction vectors into LLM's logits layer to steer LLM towards a musically interpretable concept (e.g. major vs. minor)

CoT Reasoning with Sparse Autoencoders — NOV 2024 - JAN 2025

- Devised reward function to guide LLM generation through interpretability, using clustering on SAE-based token representations

Learner, a full stack **Flask / React** app that prompts & fine-tunes LLMs to help with learning languages — SEP 2024 - DEC 2024

Convolutional Neural Net, AlexNet, UNet implementation — MAR 2024 - APR 2024

- Engineered CNN, AlexNet, UNet and variants, including construction of layers, activations, autodiff in NumPy, PyTorch

Skills

Python, PyTorch, TensorFlow, Machine Learning, Artificial Intelligence, JupyterNotebook, C, C#, C++, SQL, JavaScript, TypeScript, React, Java, Golang, Unix, Git, Docker, AWS, Ruby on Rails, Scala, Kubernetes, RabbitMQ, GraphQL, HTML, CSS, Jenkins